Classification

➢ Objectives

To understand and implement different types of classification models.

➢ Classification

Classification is a supervised learning technique used to predict the categorical class labels of new observations based on past data.

Types of Classification:

- Binary Classification: Two classes (e.g., Yes/No)
- Multiclass Classification: More than two classes (e.g., classifying types of fruits)
- Multilabel Classification: Multiple labels per instance (e.g., tagging a photo with multiple objects)

In **multiclassification** problem, most algorithms can return a **probability score** of belonging to the **class of interest**. Like predicting type of tumor normal, benign, malignant.

Most classification methods, provides two prediction methods:

- **predict** (which returns the class) and
- returns **probabilities** for each class

Popular classification algorithm

- Naïve Bayes classifier
- Logistic regression
- Decision trees
- Random forest
- Support vector machines
- K-nearest neighbours

A sliding **cutoff** can then be used to **convert the score to a decision**.

The general approach is as follows:

- **Establish a cutoff probability** for the class of interest, above which we consider a record as belonging to that class.
- **Estimate (with any model) the probability** that a record belongs to the class of interest.
- If that **probability is above** the cutoff probability, assign the new record to the class of interest.

The **higher the cutoff**, the fewer the records predicted as 1—that is, as belonging to the class of interest.

The **lower the cutoff**, the more the records predicted as 1.

➢ Naive Bayes Classifier

Naive Bayes is the most straightforward and fast classification algorithm, which is suitable for a large chunk of data.

Naive Bayes classifier is successfully used in various applications such as spam filtering, text classification, sentiment analysis, and recommender systems.

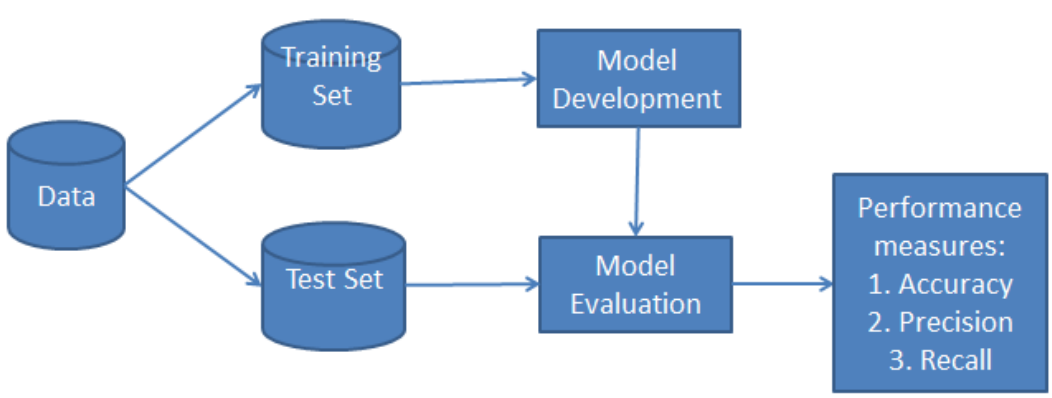
It uses Bayes theorem of probability for prediction of unknown class.

Whenever you perform classification,

- the first step is to understand the problem and identify potential features and label.
- Features are those characteristics or attributes which affect the results of the label.
- For example, in the case of a loan distribution, bank managers identify the customer's occupation, income, age, location, previous loan history, transaction history, and credit score. These characteristics are known as features that help the model classify customers.

The classification has two phases,

- a learning phase and the evaluation phase. In the learning phase, the classifier trains its model on a given dataset,
- and in the evaluation phase, it tests the classifier's performance. Performance is evaluated on the basis of various parameters such as accuracy, error, precision, and recall.



Naive Bayes classifier is the fast, accurate and reliable algorithm.

Naive Bayes classifiers have high accuracy and speed on large datasets.

Naive Bayes classifier assumes that the effect of a particular feature in a class is independent of other features.

- For example, a loan applicant is desirable or not depending on his/her income, previous loan and transaction history, age, and location. Even if these features are interdependent, these features are still considered independently.
- This assumption simplifies computation, and that's why it is considered as naive. This assumption is called class conditional independence.

Bayes' theorem is also known as **Bayes' Rule** or Bayes' law, which is used to determine the **probability of a hypothesis** with prior knowledge. It depends on the **conditional probability**.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

P(A|B) is **Posterior** probability: Probability of hypothesis A on the observed event B.

P(B|A) is **Likelihood** probability: Probability of the evidence given that the probability of a hypothesis is true.

P(A) is **Prior** Probability: Probability of hypothesis before observing the evidence.

P(B) is **Marginal** Probability: Probability of Evidence.

Naive Bayes classifier calculates the **probability of an event** in the following steps:

Step 1: Calculate the **prior probability** for given class labels

Step 2: Find **Likelihood probability** with each attribute for each class

Step 3: Put these value in Bayes Formula and **calculate posterior probability**.

Step 4: See which class has a **higher probability**, given the input belongs to the higher probability class.

For simplifying prior and posterior probability calculation, you can use the two tables **frequency and likelihood tables**.

Both of these tables will help you to calculate the prior and posterior probability. The Frequency table contains the occurrence of labels for all features.

There are two likelihood tables. Likelihood Table 1 is showing prior probabilities of labels and Likelihood Table 2 is showing the posterior probability.

Whether	Play
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	No
Sunny	Yes
Rainy	Yes
Sunny	Yes
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table

Whether	No	Yes
Overcast		4
Sunny	2	3
Rainy	3	2
Total	5	9

Likelihood Table 1

Whether	No	Yes		
Overcast		4	=4/14	0.29
Sunny	2	3	=5/14	0.36
Rainy	3	2	=5/14	0.36
Total	5	9		
	=5/14	=9/14		
	0.36	0.64		

Likelihood Table 2

Whether	No	Yes	Posterior Probability for No	Posterior Probability for Yes
Overcast		4	0/5=0	4/9=0.44
Sunny	2	3	2/5=0.4	3/9=0.33
Rainy	3	2	3/5=0.6	2/9=0.22
Total	5	9		

Probability of playing:

$$P(Yes \mid Overcast) = \frac{P(Overcast \mid Yes)\, P(Yes)}{P\,(Overcast)} \text{ .....(1)}$$

Calculate Prior Probabilities:

P(Overcast) = 4/14 = 0.29

P(Yes)= 9/14 = 0.64

Calculate **Posterior Probabilities**:

P(Overcast |Yes) = 4/9 = 0.44

Put Prior and Posterior probabilities in equation (1)

P (Yes | Overcast) = 0.44 * 0.64 / 0.29 = 0.98(Higher)

Similarly, calculate the probability of not playing:

Probability of not playing:

$$P(No \mid Overcast) = \frac{P(Overcast \mid No)\, P(No)}{P\,(Overcast)} \text{ .....(2)}$$

Calculate Prior Probabilities:

P(Overcast) = 4/14 = 0.29

P(No)= 5/14 = 0.36

Calculate **Posterior Probabilities**:

P(Overcast |No) = 0/9 = 0

Put Prior and Posterior probabilities in equation (2)

P (No | Overcast) = 0 * 0.36 / 0.29 = 0

The probability of a **'Yes' class is higher**. So you can determine here if the weather is overcast than players will play the sport.

Naive Bayes Classifier with multiple features

Now suppose you want to calculate the probability of when the weather is overcast, and the temperature is mild.

Probability of playing

Bayes Naïve Theorem says

$$P\left(Y|X\right) = \frac{P(X|Y)P(Y)}{P(X)}$$

$$P\left(Y|X_1, X_2, X_3, ..., X_n\right) = \frac{P(X_1|Y)P(X_2|Y)P(X_3|Y)...P(X_n|Y)P(Y)}{P(X_1)P(X_2)P(X_3)...P(X_n)}$$

$$P\left(Play = Yes|Weather = Overcast, Temp = Mild\right) = \frac{P\left(Weather = Overcast, Temp = Mild|Play = Yes\right)}{P(Weather = Overcast)P(Temp = Mild)} .....(1)$$

$$P\left(Play = Yes|Weather = Overcast, Temp = Mild\right) = \frac{P(Weather = Overcast|Play = Yes)P(Temp = Mild|Play = Yes)P(Play = Yes)}{P(Weather = Overcast)P(Temp = Mild)} ...(2)$$

Calculate Prior Probability: P(Yes)= 9/14 = 0.64

Calculate Likelihood Probability: P(Overcast |Yes) = 4/9 = 0.44; P(Mild |Yes) = 4/9 = 0.44

Calculate Marginal Probability: P(Overcast) = 4/14 = 0.29; P(Mild) = 6/14 = 0.4285

Now calculate Posterior Probability

$$P\left(Play = Yes|Weather = Overcast, Temp = Mild\right) = \frac{0.44 \times 0.44 \times 0.64}{0.29 \times 0.4285}$$

$$P\left(Play = Yes|Weather = Overcast, Temp = Mild\right) = 0.9970949$$

Temperature Type	Yes	No	P(Temp type)
Hot	**2**	**2**	**4/14 = 0.2857**
Mild	**4**	**2**	**6/14 = 0.4285**
Cool	**3**	**1**	**4/14 =0.2857**
Total	**9**	**5**	**14**

Types of Naive Bayes Classifier

There are three types of Naive Bayes Model, which are given below:

Gaussian: The Gaussian model assumes that features follow a normal distribution. This means if predictors take continuous values instead of discrete, then the model assumes that these values are sampled from the Gaussian distribution.

Multinomial: The Multinomial Naïve Bayes classifier is used when the data is multinomial distributed. It is primarily used for document classification problems, it means a particular document belongs to which category such as Sports, Politics, education, etc.

The classifier uses the frequency of words for the predictors.

Bernoulli: The Bernoulli classifier works similar to the Multinomial classifier, but the predictor variables are the independent Booleans variables. Such as if a particular word is present or not in a document. This model is also famous for document classification tasks.

Discriminant Analysis

Discriminant analysis is the earliest statistical classifier. It was introduced by R. A. Fisher in 1936. While discriminant analysis encompasses several techniques, the most commonly used is linear discriminant analysis, or LDA.

It have many other applications like used in principal component analysis (PCA).

Linear Discriminant Analysis (LDA) is a **supervised learning** algorithm used for **classification tasks** in machine learning. It is a technique used to f**ind a linear combination of features** that best **separates the classes** in a dataset.

LDA works by projecting the data onto a **lower-dimensional space** that **maximizes the separation between the classes**. It does this by finding a set of linear discriminants that **maximize the ratio of between-class variance to within-class variance**. In other words, it **finds the directions** in the feature space that **best separate the different classes** of data.

LDA **assumes** that the data has a **Gaussian distribution**. It also assumes that the data is **linearly separable**, meaning that a linear decision boundary can accurately classify the different classes.

To understand discriminant analysis, it is first necessary to introduce the concept of **covariance** between two or more variables.

Covariance Matrix

The **covariance** measures the relationship between two variables x and z.

Denote the mean for each variable by $\bar{X}$ and $\bar{Y}$

The covariance **Sx,z** between x and z is given by:

$$s_{x,z} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(z_i - \bar{z})}{n-1}$$

As with the **correlation coefficient**, positive values indicate a positive relationship and negative values indicate a negative relationship.

Correlation, however, is **constrained to be between –1 and 1**, whereas **covariance scale depends** on the scale of the **variables x and z**.

The **covariance matrix Σ** for x and z consists of the **individual variable variances,** S_x^2 s S_z^2, on the diagonal (where row and column are the same variable) and the covariances between variable pairs on the off-diagonals:

$$\hat{\Sigma} = \begin{bmatrix} s_x^2 & s_{x,z} \\ s_{z,x} & s_z^2 \end{bmatrix}$$

Fisher's Linear Discriminant

Fisher's linear discriminant distinguishes variation between groups, on the one hand, from variation within groups on the other.

Divides the records into two groups, linear discriminant analysis (LDA) focuses on maximizing the "between" sum of squares SSbetween (measuring the variation between the two groups) relative to the "within" sum of squares SSwithin (measuring the within-group variation).

$$\boxed{\frac{SS_{between}}{SS_{within}}}$$

LDA projects data from a D dimensional feature space down to a D' (D>D') dimensional space in a way to maximize the variability between the classes and reducing the variability within the classes.

PERFORMANCE METRICS

1. **Accuracy**

Accuracy is a fundamental metric used for evaluating the performance of a classification model. It tells us the proportion of correct predictions made by the model out of all predictions.

$$\text{Accuracy} = \frac{\text{NumberofCorrectPredictions}}{\text{TotalNumberofPredictions}}$$

While accuracy provides a quick snapshot, it can be misleading in cases of imbalanced datasets. For example, in a dataset with 90% class A and 10% class B, a model predicting only class A will still achieve 90% accuracy but it will fail to identify any class B instances.

Accuracy is good but it gives a False Positive sense of achieving high accuracy. The problem arises due to the possibility of misclassification of minor class samples being very high.

2. **Precision**

It measures how many of the positive predictions made by the model are actually correct. It's useful when the cost of false positives is high such as in medical diagnoses where predicting a disease when it’s not present can have serious consequences.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Where:

- TP = True Positives
- FP = False Positives

Precision helps ensure that when the model predicts a positive outcome, it’s likely to be correct.

3. Recall

Recall or Sensitivity measures how many of the actual positive cases were correctly identified by the model. It is important when missing a positive case (false negative) is more costly than false positives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Where:

- FN = False Negatives

In scenarios where catching all positive cases is important (like disease detection), recall is a key metric.

4. F1 Score

The F1 Score is the harmonic mean of **precision** and **recall**. It is useful when we need a balance between precision and recall as it combines both into a single number. A high F1 score means the model performs well on both metrics. Its range is [0,1].

Lower recall and higher precision gives us great accuracy but then it misses a large number of instances. More the F1 score better will be performance. It can be expressed mathematically in this way:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5. Confusion Matrix

Confusion matrix creates a N X N matrix, where N is the number of classes or categories that are to be predicted. Here we have N = 2, so we get a 2 X 2 matrix. Suppose there is a problem with our practice which is a binary classification. Samples of that classification belong to either Yes or No. So, we build our classifier which will predict the class for the new input sample. After that, we tested our model with 165 samples and we get the following result.

n=165	Predicted No	Predicted Yes
Actual No	50	10
Actual Yes	5	100

There are 4 terms we should keep in mind:

1. **True Positives:** It is the case where we predicted Yes and the real output was also Yes.
6. **True Negatives:** It is the case where we predicted No and the real output was also No.
3. **False Positives:** It is the case where we predicted Yes but it was actually No.
4. **False Negatives:** It is the case where we predicted No but it was actually Yes.